

Process Safety Management (14 Elements of PSM)

14 Elements of Process Safety Management

The Process Safety and Risk Management Model

OSHA mandates that employers have a Process Safety Management (PSM) system in place that follows these 14 rules and practices.

HHC = Highly Hazardous Chemicals



What is Process Safety Management?

Process Safety Management (PSM) is an Occupational Safety and Health Administration (OSHA) regulatory standard designed for processes involving Highly Hazardous Chemicals (HHCs). The OSHA process safety management standard includes requirements for preventing catastrophic releases of toxic, reactive, flammable, or explosive chemicals. While the full list of HHCs can be found in [Appendix A](#) of the OSHA PSM standard, it might also be helpful for chemical safety managers to know the basic definitions of toxic, reactive, flammable, and explosive. These are properties or characteristics of HHCs.

According to the Center for Chemical Process Safety (CCPS):

- **Toxic Material** – an airborne agent that could result in acute adverse human health effects
- **Reactive Material** – a substance which enters into a chemical reaction with other stable or unstable material
- **Flammable** – a gas that can burn with a flame if mixed with a gaseous oxidizer such as air and then ignited
- **Explosive** – a chemical that causes a sudden release of pressure, gas, and heat when subjected to sudden shock, pressure, or high temperature

Why Is Process Safety Management Important?

One of the reasons why OSHA decided to create PSM is the Phillips disaster of 1989. In October of that year, an explosion and fire at the Phillips 66 Company's Houston Chemical Complex (HCC) took the lives of 23 workers and injured 132 others. The event also resulted in \$750,000,000 worth of property damage and a \$4,000,000 fine issued by OSHA.

However, even after OSHA had released the PSM standard in 1992, another disaster occurred at the BP Texas City refinery in 2005, leading to the deaths of 15 workers and 180 others injured. The company had to pay \$2,000,000,000 to the victims of the disaster and their relatives as compensation. Repairs and lost profits also cost BP over \$1,000,000,000.

What Falls Under the PSM Standard and What Doesn't

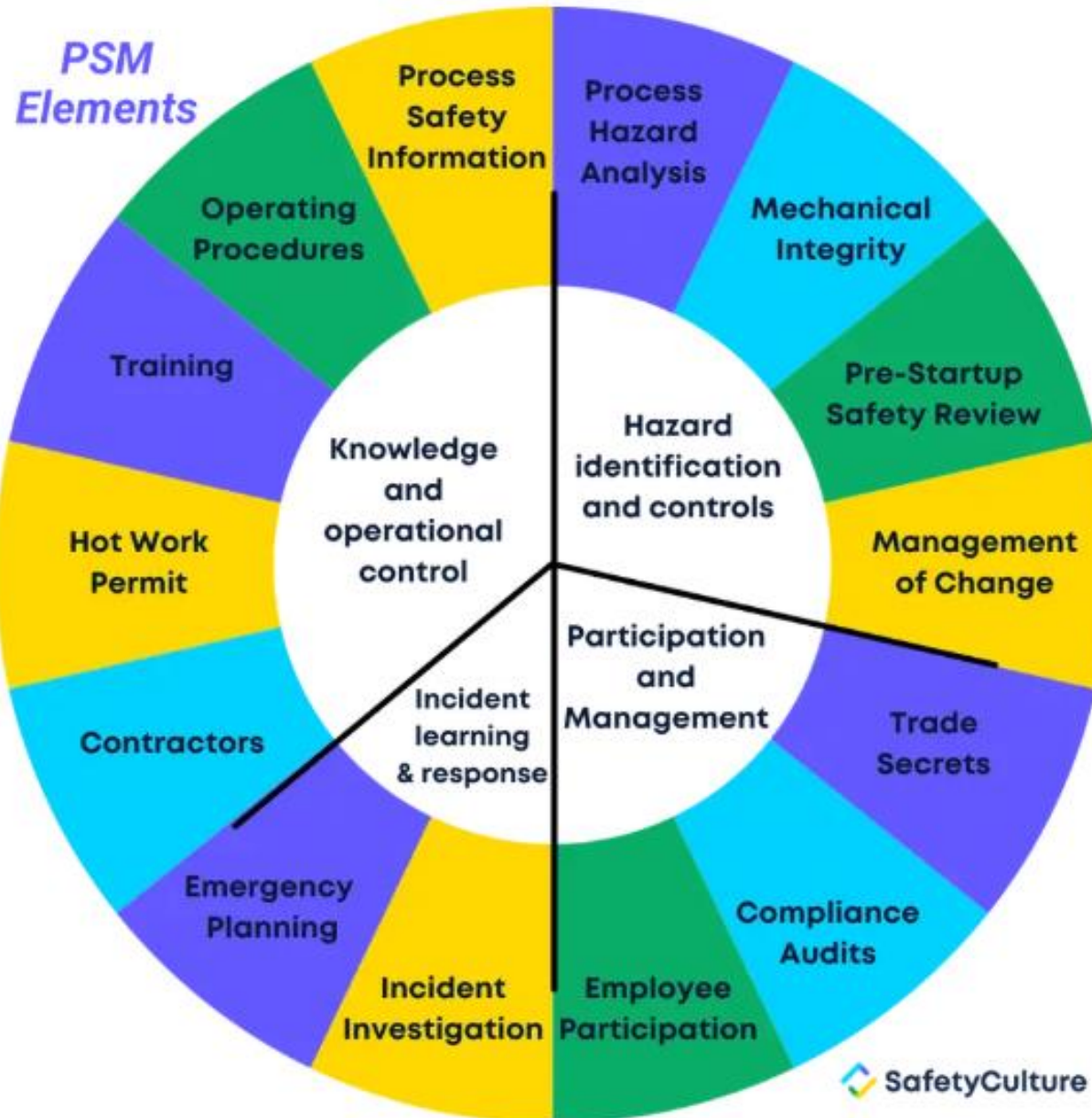
OSHA states that the [process safety management standard](#) applies to any process that involves a chemical with quantities at or above the specified threshold listed in [Appendix A](#) and any process that involves a [Category 1 flammable gas](#) or a flammable liquid with a [flashpoint](#) below 100 °F (37.8 °C) and in a quantity of 10,000 pounds (4535.9 kg) or more.

Compliance with PSM is not required when hydrocarbon fuels are used solely for workplace consumption and when flammable liquids are kept below their normal boiling point. PSM also does not apply to retail facilities, oil or gas well drilling or servicing operations, and normally unoccupied remote facilities.

The 14 PSM Elements

For organizations with processes that involve the use of HHCs, [complying with process safety management](#) is crucial, not only because it's a legal requirement, but also because the consequences of non-compliance are devastating.

To prevent disasters such as the Phillips 1989 and the BP Texas City Refinery explosions, chemical safety managers need to ensure that each site has a process safety management program with all of the 14 elements.



1. Process Safety Information

Collect and document process safety information (PSI) regarding hazards, technology, and equipment. Hazard information should cover toxicity, exposure limits, physical and chemical properties, reactivity, stability, and risks of mixing materials. If [safety data sheets](#) comply with [OSHA standards](#), hazard information is considered met. Technology information should include diagrams, chemistry details, inventory limits, and consequences of deviations. Equipment information entails construction materials, diagrams, safety systems, and design standards.

2. Operating Procedures

Develop and implement operating procedures on how to safely proceed with the activities and tasks involved in the process. Operating procedures should be aligned with PSI and address the steps for each operating phase, operating limits, safety and health considerations, and safety systems and their functions.

Aside from PSI on hazards, other safety and health considerations are the precautions necessary to prevent exposure, control measures in case of exposure, quality control for raw materials, and control of hazardous chemical inventory levels.

3. Process Hazard Analysis

A process hazard analysis (PHA) evaluates process hazards, considering past incidents, control measures, and potential consequences of failures. It requires a specialized team comprising experts in engineering, process operations, and PHA methodologies. Such methodologies include PHA checklists, what-if scenarios, , HAZOP, FMEA, and fault tree analysis. PHAs, a vital part of process safety management, must be updated and revalidated every five years.

4. Employee Participation

Before proceeding with the other elements, chemical safety managers will need to develop a plan of action on implementing employee participation in the PSM program. Additionally, chemical safety managers must consult with employees on how to proceed with each element as well as provide access to information relevant to the program.

5. Training

Aside from participating in the PSM program, employees should receive adequate training on the process, operating procedures, emergency operations, specific safety and health hazards, and safe work practices. Chemical safety managers should also conduct refresher training at least every 3 years, or more often if necessary.

All training must be documented and include the identity of the employee, the date of training, and verification that the employee understands the subject matter of the training.

6. Contractors

This PSM element applies to contractors performing maintenance or repair, turnaround, major renovation, and specialty work. Chemical safety managers should review the contractor's safety performance and programs before selecting them.

After selection, chemical safety managers must periodically evaluate the performance of the chosen contractor. They should also consider giving the contractor's employees a [site induction](#) to brief them on the potential hazards related to the work, process, or site.

7. Mechanical Integrity

Ensure the mechanical integrity of process equipment, including [pressure vessels](#), [storage tanks](#), [piping systems](#), [pumps](#), and monitoring systems. Document inspections and tests conducted, rectifying any deficiencies beyond acceptable limits before further use. Utilize [tracking tools](#) or [software](#) to monitor equipment status, promptly addressing any issues to prevent accidents.

8. Incident Investigation

A PSM incident involves or could potentially result in a catastrophic release of a hazardous chemical. Chemical safety managers must begin an investigation within 48 hours of such an incident, identifying its primary cause and contributing factors, along with recommendations for prevention. Prompt action should follow to address findings and implement corrective measures from the investigation report. Once resolved, all affected parties should review the report thoroughly to determine the need for further investigation.

9. Management of Change

Before making any changes to chemicals, technology, equipment, or facilities, prepare a [management of change \(MOC\)](#) procedure. It should address the technical basis for the proposed change, its impact on safety and health, modifications to operating procedures, and the authorization requirements and time period needed for the proposed change.

10. Pre-Startup Safety Review

For new and modified facilities, chemical safety managers must perform a pre-startup safety review. The [PSSR](#) should confirm that construction and equipment are in accordance with design specifications, operating procedures have been established, and all employee training is complete. Additionally, there must be a MOC procedure for modified facilities and a PHA has to be performed for new facilities.

11. Emergency Planning and Response

Develop an emergency action plan ([EAP](#)) for the entire site. It should meet the requirements of the [EAP OSHA standard](#) and include procedures for handling small releases of HHCs. If the site is under the Hazardous Waste Operations and Emergency Response ([HAZWOPER](#)) [OSHA standard](#), ensure that the EAP meets the requirements of that standard as well.

12. Hot Work Permit

A [hot work permit](#) is issued whenever hot work operations are conducted on or near a process. Before requesting or issuing a hot work permit, chemical safety managers must ensure that the fire prevention and protection requirements of the [Welding, Cutting, and Brazing OSHA standard](#) have been implemented. The hot work permit will verify this and also specify the date/s authorized for hot work and the object/s on which hot work is to be performed.

13. Compliance Audits

Chemical safety managers must conduct [compliance audits](#) at least every 3 years to check if the site's PSM program adheres to all of the elements. They will also need to determine and document an appropriate response to each of the findings in the compliance audit reports. Organizations are required by the PSM standard to keep the two most recent compliance audit reports.

14. Trade Secrets

Without regard to possible trade secret status, all information necessary to comply with the PSM standard should be made available to those involved in the PSM program. However, organizations are allowed to require such persons to enter into confidentiality agreements not to disclose any trade secret information.

Prevent Disastrous Decisions

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Analyze past decisions that led to incidents to avoid similar disasters and ensure positive outcomes of future critical decisions.

Examples of critical decisions

Critical decisions can either prevent or contribute to disastrous outcomes. In many chemical incidents, unfortunately, critical decisions have been a contributing factor.

Macondo well blowout and explosion. The Deepwater Horizon drilling rig explosion and subsequent fire occurred on April 20, 2010 (1). The rig was drilling in the Macondo Prospect oil field about 40 mi (64 km) off the southeast coast of Louisiana. The explosion and subsequent fire sunk the Deepwater Horizon rig, killing 11 workers and injuring 17 others (Figure 1) (1). The blowout also caused an

oil-well fire and a massive offshore oil spill of more than 4 million bbl in the Gulf of Mexico. The incident has been described as the largest accidental marine oil spill in the world and the largest environmental disaster in U.S. history.

Prior to the explosion, an engineer decided not to test the integrity of a very deep offshore well and canceled the integrity test (a cement bond log test), going against the approved work plan and established industry practices. As a result, methane gas that had been leaking into the well moved up the pipe to the rig, where it ignited.



▲ **Figure 1.** An engineer made a disastrous decision to not test the integrity of a deep offshore well, which contributed to the largest environmental disaster in the U.S. — the Macondo well blowout incident. Source: (1).

Richmond refinery fire. On Aug. 6, 2012, a pipe in a crude unit atmospheric column catastrophically ruptured at the Chevron refinery in Richmond, CA (2,3). The pipe rupture occurred on a 52-in.-long section of an 8-in.-dia. line. At the time of the incident, light gas oil was flowing through the line at a rate of approximately 10,800 bbl/day. Six employees suffered minor injuries during the incident and subsequent emergency response efforts. In the weeks after the incident, nearly 15,000 people from the surrounding communities sought medical treatment at nearby medical facilities for various respiratory ailments, and approximately 20 of those people were admitted for treatment.

Prior to this incident, a plant manager ignored expert advice to replace piping on a product takeoff line in a crude distillation unit that had been severely corroded by a well-known mechanism. The same plant manager continued operations when leaks from the corroded piping were noticed, exposing emergency crews to unnecessary risk.



▲ **Figure 2.** A risk-assessment team violated company risk-management procedures and siting criteria when they allowed temporary office trailers to be placed in close proximity to hydrocarbon processing units that were temporarily offline. During startup of the units, an incident occurred that killed 15 workers in the nearby office trailers. Source: (4).

Texas City refinery explosion. On March 23, 2005, a hydrocarbon vapor cloud formed in an isomerization process unit that was starting up after a shutdown at BP's refinery in Texas City, TX (4–6). The vapor cloud ignited and violently exploded, killing 15 workers, injuring 180 others, and severely damaging the refinery (Figure 2) (4).

Long before the explosion occurred, a risk-assessment team approved the siting of temporary office facilities in close proximity to hydrocarbon processing units that were temporarily offline, violating the company's risk procedures and facility siting criteria. All of the fatalities in this incident occurred in or near the temporary office facilities.

Longford gas plant explosion. On Sept. 25, 1998, the heated lean oil system in a heat exchanger failed at a gas plant (7). Without the flow of heated oil, low temperatures caused cold temperature embrittlement in the heat exchanger. When heated oil was reintroduced into the heat exchanger, a brittle fracture released a large cloud of hydrocarbon vapors. The vapor cloud ignited (without causing an explosion) and set off a fierce jet fire that lasted for two days.

In this incident, the company's engineering department — which was responsible for maintaining corporate safety standards and good engineering practices — failed to conduct a hazard and operability (HAZOP) analysis of the changes made to the gas plant.

Bhopal chemical plant release. A major gas leak occurred at the Union Carbide India Ltd. pesticide plant in Bhopal, India, on Dec. 2, 1984 (8). Highly toxic methyl isocyanate gas spread to nearby small towns and affected over half a million people. The official immediate death toll was 2,259, but experts estimate the total number of deaths to be much higher. Union Carbide Corp., a major international chemical company, compromised its corporate safety standards and good engineering practices when operating the Bhopal chemical plant.



▲ **Figure 3.** A good decision was made to delay the startup of the Terra Nova floating production storage and offloading (FPSO) vessel when a pressure balance safety joint unexpectedly failed, avoiding a disastrous oil spill. Image courtesy of Terra Nova Alliance.

Terra Nova offshore oil platform. The Terra Nova floating production storage and offloading (FPSO) vessel is located in the Terra Nova oil and gas field, approximately 220 mi (350 km) off the coast of Newfoundland, Canada, in the North Atlantic Ocean (Figure 3). It produces oil from subsea wells that are connected to the vessel with flexible flowlines or risers.

A senior vice president of operations delayed startup of the multibillion-dollar offshore oil production facility when a pressure balance safety joint failed on a flow line. His decision to delay oil production prevented the company from receiving a significant tax credit, but ignoring the failure and proceeding with startup could have caused a disastrous oil spill.



▲ **Figure 4.** A potentially disastrous incident was avoided at one of the largest energy-related ventures in the Middle East — the Dolphin Energy gas project. Management made a critical decision to stop production when a minor leak developed in an overhead line. Image courtesy of Dolphin Energy.

Dolphin Energy gas project. The Dolphin Energy gas project began operations in 2007 to recover gas reserves off the coast of Qatar. The recovered gas is transported to an onshore gas processing plant. After processing, the gas is transported through a 48-in. subsea pipeline to the United Arab Emirates and then to Oman. The Dolphin Energy project is one of the largest energy-related ventures operating in the region.

A vice president of operations stopped production — risking significant penalties for not achieving annual production targets — when an overhead line from the condensate stabilization unit developed a leak (Figure 4), threatening workers with exposure to high levels of hydrogen sulfide. Operations resumed at reduced capacity and pressure only after a risk assessment was conducted and interim mitigation measures were identified and implemented.

To determine the most significant factors that have historically played a role in safety incidents, we evaluated eleven past disasters (1–14):

- Flixborough plant explosion (June 1, 1974)
- Bhopal chemical plant release (Dec. 3, 1984)
- Piper Alpha oil platform explosion (July 6, 1988)
- Pasadena polyethylene plant explosion (Oct. 23, 1989)
- Longford gas plant explosion (Sept. 25, 1998)
- Petrobras 36 (P-36) oil platform explosion

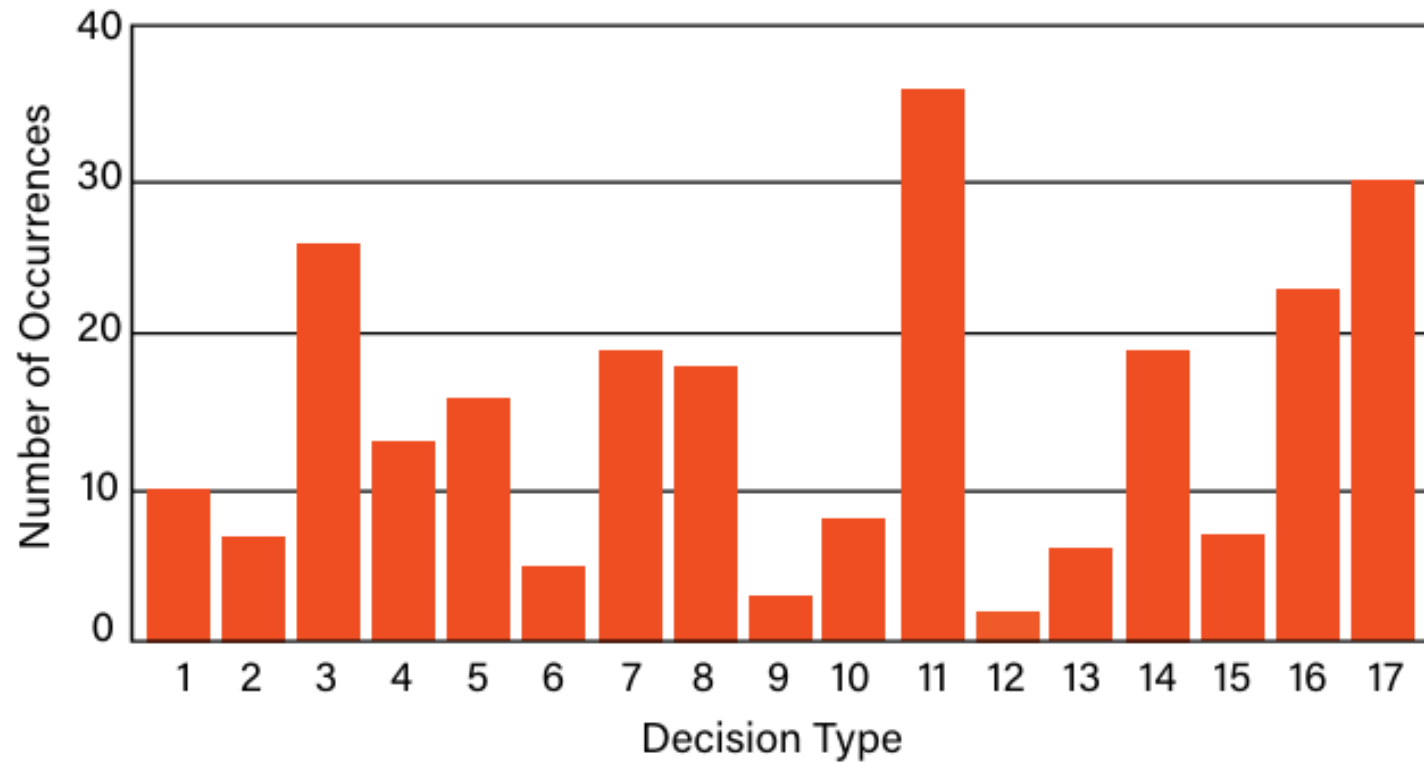
(March 20, 2001)

- Texas City refinery explosion (March 23, 2005)
- Point Comfort propylene explosion (Oct. 6, 2005)
- Macondo well blowout and explosion (April 20, 2010)
- Richmond refinery fire (Aug. 6, 2012)
- Geismar olefins plant explosion and fire

(June 13, 2013).

Table 1. Identifying the common themes of chemical incidents helps to inform better decisions and prevent similar disastrous outcomes.

Decision Type	Decision Theme
1	Significantly changing the work execution plans without approvals
2	Making impactful decisions that are well beyond one's responsibilities
3	Not conducting revalidation risk assessments (<i>i.e.</i> , failure to capture changing circumstances and situational creep)
4	Ignoring the advice of subject matter experts
5	Not following established management of change (MOC) processes
6	Keeping safety-critical equipment offline longer than necessary
7	Practicing poor control of work and handover management
8	Failing to recognize and respond effectively to emergency situations, including not having emergency response (ER) plans and not conducting ER exercises
9	Poorly managing incentive programs
10	Not maintaining robust communication protocols
11	Not following established engineering standards and norms
12	Not ensuring that all third-party-supplied safety-critical equipment is performance tested and verified prior to use
13	Failing to inform senior management (<i>i.e.</i> , preventing bad news from reaching upper management)
14	Cost-cutting without assessment of impacts on operations and staff
15	Failing to learn from recent similar incidents
16	Being pressured to maintain production at the expense of safety
17	Lacking adequate technical operational expertise and/or lacking sufficient training



▲ **Figure 5.** Learning from reoccurring disastrous decisions is a practical approach to improving critical decisions. The most common types of decisions in our incident analyses were not following established engineering standards and norms, lacking adequate technical operational expertise and/or lacking sufficient training, and not conducting revalidation risk assessments.

Table 2. An incident decision matrix is a useful tool for analyzing the key decisions that directly or indirectly caused an incident. This incident decision matrix summarizes the decisions that contributed to the Macondo well blowout.

Disastrous Decision	Decision Significance	Rationale	Factors that Contributed Over Time	Immediate Impact	Improved Decision	Decision Type
Dismissing Halliburton's recommendation to use 21 centralizers to ensure that the production casing ran down the center of the well bore to minimize channel formation	Significant	Avoid delays and extra costs	The cement slurry failed to rise uniformly between the casing and the borehole wall, creating channels (or spaces) that can contribute to a well blowout	None — the impact of channeling would only come into play during a well incident	Install the number of centralizers that was recommended by Halliburton and confirmed by models	1
Choosing a single-well design over a tie-back design, which had more barriers to prevent gas from flowing up to the outside of the well	Significant	Reduce costs	High-pressure gas that entered the bottom of the well was able to escape to the outside of the well more easily	None — the impact of channeling would only come into play during a well incident	Invest in the robust tie-back design that provides extra layers of protection	1

Rewarding production and cost reduction while not incentivizing safety, creating pressure to work fast and take risks	Significant	Increase production	Risk taking was incentivized over safety	None — the company culture of cutting corners persisted and eventually had a direct impact on the disaster	Provide balanced incentives for both safety and cost reduction goals Conduct risk assessments to ensure that cost reduction incentives do not adversely affect safety	7
Initiating the cement process without reviewing the cement design or waiting for test results	Critical	Avoid delays	Important steps were ignored to stay on schedule	The unstable cement was a significant factor in the well blowout and caused the disaster	Follow the established plan and wait for test results before beginning critical work	1

Table 3. This incident decision matrix breaks down the decisions that contributed to the Longford gas explosion.

Disastrous Decision	Decision Significance	Rationale	Factors that Contributed Over Time	Immediate Impact	Improved Decision	Decision Type
Ignoring engineering standards and practices and failing to conduct HAZOP studies at the gas plant facility	Important	Disregard for the importance of risk assessments	Hazards related to maintaining the temperature profile in the de-ethanizer column were not understood	None — it was not until a major upset occurred that a lack of understanding of the tower cooling effects was revealed, directly causing the disaster	Follow engineering standards and practices, which includes conducting a HAZOP study to identify hazards and operational concerns	5, 11
Relocating all engineering staff from the site to Melbourne, which was several hundred miles away	Important	Reduce costs	Engineers lost day-to-day familiarization with the operation over time, and onsite engineering support and expertise were unavailable	None — it was not until a major upset occurred that a lack of understanding of the tower cooling effects was revealed, directly causing the disaster	Assign a core staff of process engineers on-site to support operations and to conduct risk assessments	14

Poorly training personnel and failing to address hazardous scenarios	Important	Reduce costs	Operators were unaware of the critical need to prevent the rich oil system from reaching low temperatures during an upset	None — it was not until a major upset occurred that a lack of understanding of the tower cooling effects was revealed, directly causing the disaster	Provide incident-based training for all operators	4, 14, 17
Not identifying significant issues, such as a lack of risk assessments, with an effective audit program	Important	Change in audit program	Plant personnel were under the impression that procedures and safety were satisfactory	None — it was not until a major upset occurred that the lack of feedback from audits was revealed	Assign experienced technical members to the audit team with the task of identifying areas of improvement	15
Not conducting detailed investigations of incidents that occurred	Important	Reduce costs	Opportunities were not available to learn from process upsets and standard operating procedures (SOPs) were not updated	None — it was not until a major upset occurred that a lack of understanding of the tower cooling effects was revealed, directly causing the disaster	Conduct thorough investigations of all incidents at Longford and other similar operations Update SOPs as warranted	15, 17

Decision types. The incident analyses indicated that 17 different types of decisions (Table 1) are common in chemical disasters. Avoiding and learning from these — especially the most frequently occurring ones (Figure 5) — can significantly improve the outcomes of critical decisions.

The most important types of decisions to learn from and avoid are:

- not following established engineering standards and norms (Type 11)
- lacking adequate technical operational expertise and/or lacking sufficient training (Type 17)
- not conducting revalidation risk assessments (*i.e.*, failure to capture changing circumstances and situational creep) (Type 3).

These would likely be the most frequent types of critical decisions regardless of the number of disastrous incidents included in the analysis.